

Brascamp–Lieb Inequalities, Young’s Inequality, and Optimal Transport

1 The Brascamp–Lieb inequality

The Brascamp–Lieb inequality is a common generalisation of several classical integral inequalities, including Hölder’s inequality, the Loomis–Whitney inequality, and Young’s convolution inequality.

Definition 1 (Brascamp–Lieb datum). *A Brascamp–Lieb datum consists of surjective linear maps*

$$B_j : \mathbb{R}^n \longrightarrow \mathbb{R}^{n_j}, \quad j = 1, \dots, m,$$

together with non-negative exponents

$$p_j \geq 0.$$

For non-negative integrable functions

$$f_j : \mathbb{R}^{n_j} \rightarrow [0, \infty),$$

the Brascamp–Lieb functional is

$$\mathcal{B}_{\mathbf{B}, \mathbf{p}}(f_1, \dots, f_m) := \frac{\int_{\mathbb{R}^n} \prod_{j=1}^m f_j(B_j x)^{p_j} dx}{\prod_{j=1}^m \left(\int_{\mathbb{R}^{n_j}} f_j \right)^{p_j}}.$$

The optimal Brascamp–Lieb constant is

$$\mathrm{BL}(\mathbf{B}, \mathbf{p}) := \sup_{f_j \geq 0} \mathcal{B}_{\mathbf{B}, \mathbf{p}}(f_1, \dots, f_m).$$

Thus the Brascamp–Lieb inequality is

$$\int_{\mathbb{R}^n} \prod_{j=1}^m f_j(B_j x)^{p_j} dx \leq \mathrm{BL}(\mathbf{B}, \mathbf{p}) \prod_{j=1}^m \left(\int_{\mathbb{R}^{n_j}} f_j \right)^{p_j}.$$

The finiteness of the constant is governed by two geometric conditions. The first is the scaling condition

$$n = \sum_{j=1}^m p_j n_j.$$

This is necessary by dilating all the functions simultaneously. The second is the dimension condition

$$\dim V \leq \sum_{j=1}^m p_j \dim B_j V \quad \text{for every subspace } V \leq \mathbb{R}^n.$$

This prevents mass from concentrating along a subspace on which the right-hand side has insufficient dimension to control the left-hand side.

Theorem 1 (Finiteness criterion). *For a non-degenerate Brascamp–Lieb datum, the constant $\text{BL}(\mathbf{B}, \mathbf{p})$ is finite if and only if*

$$n = \sum_{j=1}^m p_j n_j$$

and

$$\dim V \leq \sum_{j=1}^m p_j \dim B_j V \quad \text{for every subspace } V \leq \mathbb{R}^n.$$

2 Lieb’s Gaussian formula

A central theorem of Lieb says that the optimal Brascamp–Lieb constant is already detected by Gaussian functions.

For a positive definite matrix A_j on $\mathbb{R}^{n_j}$, define the centred Gaussian

$$G_{A_j}(y) := e^{-\pi \langle A_j y, y \rangle}.$$

Then

$$\int_{\mathbb{R}^{n_j}} G_{A_j}(y) dy = (\det A_j)^{-1/2}.$$

On the other hand,

$$\prod_{j=1}^m G_{A_j}(B_j x)^{p_j} = \exp \left(-\pi \left\langle \left(\sum_{j=1}^m p_j B_j^* A_j B_j \right) x, x \right\rangle \right).$$

Therefore

$$\int_{\mathbb{R}^n} \prod_{j=1}^m G_{A_j}(B_j x)^{p_j} dx = \det \left(\sum_{j=1}^m p_j B_j^* A_j B_j \right)^{-1/2}.$$

The value of the Brascamp–Lieb functional on these Gaussians is

$$\left[\frac{\prod_{j=1}^m (\det A_j)^{p_j}}{\det \left(\sum_{j=1}^m p_j B_j^* A_j B_j \right)} \right]^{1/2}.$$

Theorem 2 (Lieb’s Gaussian theorem). *The sharp Brascamp–Lieb constant is given by the finite-dimensional Gaussian optimisation problem*

$$\text{BL}(\mathbf{B}, \mathbf{p}) = \sup_{A_j > 0} \left[\frac{\prod_{j=1}^m (\det A_j)^{p_j}}{\det \left(\sum_{j=1}^m p_j B_j^* A_j B_j \right)} \right]^{1/2}.$$

In particular, whenever the supremum is attained by positive definite matrices A_j , the corresponding centred Gaussians

$$G_{A_j}(y) = e^{-\pi \langle A_j y, y \rangle}$$

are extremisers for the Brascamp–Lieb inequality.

In degenerate cases the supremum need not be attained, but Lieb’s theorem still says that Gaussian extremising sequences determine the optimal constant.

3 Young's inequality

Young's convolution inequality is one of the model examples contained in the Brascamp–Lieb framework.

Definition 2 (Young's convolution inequality). *Let $1 \leq p, q, r \leq \infty$ satisfy*

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}.$$

Then

$$\|F * G\|_{L^r(\mathbb{R}^n)} \leq \|F\|_{L^p(\mathbb{R}^n)} \|G\|_{L^q(\mathbb{R}^n)}.$$

The sharp form, due to Beckner and Brascamp–Lieb, is

$$\|F * G\|_{L^r(\mathbb{R}^n)} \leq \left(\frac{A_p A_q}{A_r} \right)^n \|F\|_{L^p(\mathbb{R}^n)} \|G\|_{L^q(\mathbb{R}^n)},$$

where

$$A_s := s^{1/(2s)} (s')^{-1/(2s')}, \quad \frac{1}{s} + \frac{1}{s'} = 1,$$

with the usual endpoint convention $A_1 = A_\infty = 1$.

To see Young's inequality as a Brascamp–Lieb inequality, use duality. For $1 < r < \infty$,

$$\|F * G\|_r = \sup_{\|H\|_{r'}=1} \int_{\mathbb{R}^n} (F * G)(z) H(z) dz.$$

Writing out the convolution gives the trilinear form

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x) G(y) H(x+y) dx dy.$$

This is a Brascamp–Lieb expression on the ambient space $\mathbb{R}^n \times \mathbb{R}^n$, with linear maps

$$B_1(x, y) = x, \quad B_2(x, y) = y, \quad B_3(x, y) = x + y.$$

Set

$$c_1 = \frac{1}{p}, \quad c_2 = \frac{1}{q}, \quad c_3 = \frac{1}{r'}.$$

If

$$f_1 = F^p, \quad f_2 = G^q, \quad f_3 = H^{r'},$$

then

$$f_1(B_1(x, y))^{c_1} f_2(B_2(x, y))^{c_2} f_3(B_3(x, y))^{c_3} = F(x) G(y) H(x+y).$$

The Brascamp–Lieb inequality therefore gives

$$\int_{\mathbb{R}^n} \int_{\mathbb{R}^n} F(x) G(y) H(x+y) dx dy \leq C \|F\|_p \|G\|_q \|H\|_{r'}.$$

The scaling condition for this Brascamp–Lieb datum is

$$2n = n(c_1 + c_2 + c_3),$$

which is equivalent to

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r'} = 2.$$

Since $1/r' = 1 - 1/r$, this is exactly

$$\frac{1}{p} + \frac{1}{q} = 1 + \frac{1}{r}.$$

Thus Young's inequality is precisely the Brascamp–Lieb inequality associated with the three maps

$$(x, y) \mapsto x, \quad (x, y) \mapsto y, \quad (x, y) \mapsto x + y.$$

The sharp Young constant is the Gaussian Brascamp–Lieb constant for this datum:

$$C = \left(\frac{A_p A_q}{A_r} \right)^n.$$

Equivalently,

$$C = (A_p A_q A_{r'})^n,$$

since $A_{r'} = A_r^{-1}$.

4 Optimal transport and the geometric Brascamp–Lieb inequality

The connection with optimal transport is clearest after putting the datum in geometric position.

Definition 3 (Geometric Brascamp–Lieb datum). *A Brascamp–Lieb datum is in geometric position if the maps*

$$U_j : \mathbb{R}^n \rightarrow \mathbb{R}^{n_j}$$

satisfy

$$U_j U_j^* = I_{\mathbb{R}^{n_j}}$$

and

$$\sum_{j=1}^m p_j U_j^* U_j = I_{\mathbb{R}^n}.$$

In this case the Brascamp–Lieb inequality has constant 1:

$$\int_{\mathbb{R}^n} \prod_{j=1}^m f_j(U_j x)^{p_j} dx \leq \prod_{j=1}^m \left(\int_{\mathbb{R}^{n_j}} f_j \right)^{p_j}.$$

The standard Gaussian already gives equality. Indeed, with

$$\gamma_k(x) = e^{-\pi|x|^2},$$

we have $\int_{\mathbb{R}^k} \gamma_k = 1$, and

$$\prod_{j=1}^m \gamma_{n_j}(U_j x)^{p_j} = \exp \left(-\pi \sum_{j=1}^m p_j |U_j x|^2 \right).$$

Using

$$\sum_{j=1}^m p_j U_j^* U_j = I_{\mathbb{R}^n},$$

we get

$$\sum_{j=1}^m p_j |U_j x|^2 = |x|^2.$$

Hence

$$\int_{\mathbb{R}^n} \prod_{j=1}^m \gamma_{n_j}(U_j x)^{p_j} dx = \int_{\mathbb{R}^n} e^{-\pi|x|^2} dx = 1.$$

Now let f_j be smooth positive probability densities on $\mathbb{R}^{n_j}$. Let $T_j = \nabla \phi_j$ be the Brenier map transporting $f_j(y) dy$ optimally, for quadratic cost, to the standard Gaussian measure $\gamma_{n_j}(z) dz$. The Monge–Ampere equation gives

$$f_j(y) = \gamma_{n_j}(T_j(y)) \det DT_j(y).$$

Define

$$S(x) := \sum_{j=1}^m p_j U_j^* T_j(U_j x).$$

Since each T_j is the gradient of a convex function, S is also the gradient of a convex function:

$$S = \nabla \left(\sum_{j=1}^m p_j \phi_j(U_j x) \right).$$

Its Jacobian is

$$DS(x) = \sum_{j=1}^m p_j U_j^* DT_j(U_j x) U_j.$$

Two inequalities are the heart of Barthe’s transport proof. First, by the geometric condition and Cauchy’s inequality,

$$|S(x)|^2 \leq \sum_{j=1}^m p_j |T_j(U_j x)|^2.$$

Second, by the determinant inequality associated with geometric Brascamp–Lieb data,

$$\det DS(x) \geq \prod_{j=1}^m \det DT_j(U_j x)^{p_j}.$$

Combining these with the Monge–Ampere identities yields

$$\begin{aligned} \prod_{j=1}^m f_j(U_j x)^{p_j} &= \prod_{j=1}^m \gamma_{n_j}(T_j(U_j x))^{p_j} \prod_{j=1}^m \det DT_j(U_j x)^{p_j} \\ &= \exp \left(-\pi \sum_{j=1}^m p_j |T_j(U_j x)|^2 \right) \prod_{j=1}^m \det DT_j(U_j x)^{p_j} \\ &\leq e^{-\pi|S(x)|^2} \det DS(x) \\ &= \gamma_n(S(x)) \det DS(x). \end{aligned}$$

Since S is a gradient map, the area formula gives

$$\int_{\mathbb{R}^n} \gamma_n(S(x)) \det DS(x) dx \leq \int_{\mathbb{R}^n} \gamma_n(z) dz = 1.$$

Therefore

$$\int_{\mathbb{R}^n} \prod_{j=1}^m f_j(U_j x)^{p_j} dx \leq 1.$$

By homogeneity this gives the full geometric Brascamp–Lieb inequality.

5 From the general case to the geometric case

Suppose the Gaussian optimisation problem is attained by positive definite matrices A_j , and set

$$M := \sum_{j=1}^m p_j B_j^* A_j B_j.$$

Define

$$U_j := A_j^{1/2} B_j M^{-1/2}.$$

Then

$$\sum_{j=1}^m p_j U_j^* U_j = I_{\mathbb{R}^n}.$$

Moreover, the Euler–Lagrange equations for the Gaussian determinant problem give

$$A_j^{-1} = B_j M^{-1} B_j^*.$$

Consequently,

$$U_j U_j^* = A_j^{1/2} B_j M^{-1} B_j^* A_j^{1/2} = I_{\mathbb{R}^{n_j}}.$$

Thus the Gaussian extremiser puts the datum into geometric position. In that position, the standard Gaussian extremises the inequality with constant 1. Undoing the linear changes of variables gives the extremising Gaussians

$$G_{A_j}(y) = e^{-\pi \langle A_j y, y \rangle}$$

for the original datum.

6 Why Gaussians are maximisers

There are three complementary ways to understand the role of Gaussians.

First, Lieb’s theorem gives the exact statement:

$$\text{BL}(\mathbf{B}, \mathbf{p}) = \sup_{\text{centred Gaussians}} \mathcal{B}_{\mathbf{B}, \mathbf{p}}.$$

Thus no non-Gaussian family of functions can produce a larger value than the best Gaussian family.

Second, Gaussians are algebraically stable under the operations appearing in Brascamp–Lieb inequalities. If

$$f_j(y) = e^{-\pi \langle A_j y, y \rangle},$$

then

$$\prod_{j=1}^m f_j(B_j x)^{p_j}$$

is again a Gaussian in x , because the exponent is a quadratic form:

$$\sum_{j=1}^m p_j \langle A_j B_j x, B_j x \rangle = \left\langle \left(\sum_{j=1}^m p_j B_j^* A_j B_j \right) x, x \right\rangle.$$

Therefore the infinite-dimensional variational problem over all functions collapses, for Gaussians, to a finite-dimensional determinant problem.

Third, the optimal-transport proof explains why Gaussians are the equality case. In geometric position, arbitrary densities f_j are transported to Gaussians by Brenier maps T_j . The proof compares the original integral to

$$\int_{\mathbb{R}^n} \gamma_n(S(x)) \det DS(x) dx.$$

The comparison uses two inequalities: a quadratic Cauchy inequality and a determinant inequality. When the f_j are Gaussians, the Brenier maps are linear; in geometric position they are simply the identity maps. Then both inequalities become equalities. For genuinely non-Gaussian inputs, the Brenier maps are nonlinear, and the determinant and quadratic steps create loss, except in cases forced by the symmetries or factorisations of the datum.

For Young's inequality this recovers the classical sharp statement: in the strict range $1 < p, q, r < \infty$, the extremisers for the sharp Young inequality are Gaussian functions, up to the natural symmetries of translation, dilation, multiplication by constants, and common linear changes of variables.

7 Entropy viewpoint

There is also an entropy formulation. Let X be a random vector in $\mathbb{R}^n$ with density ρ , and let $B_j X$ have density ρ_j . The Brascamp–Lieb inequality is dual to the entropy inequality

$$h(X) \leq \sum_{j=1}^m p_j h(B_j X) + \log \text{BL}(\mathbf{B}, \mathbf{p}),$$

where

$$h(X) = - \int_{\mathbb{R}^n} \rho(x) \log \rho(x) dx$$

is differential entropy. This is another place where Gaussians naturally enter: among all random variables with fixed covariance, Gaussians maximise entropy. Thus the Gaussian extremisers are consistent with both the determinant formula and the transport/entropy interpretation.

References

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